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$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\ln(1 + 2 \sin x)}{\sin x} \\ & \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{2 \cos x}{1 + 2 \sin x}}{\cos x} \\ & = \lim_{x \rightarrow 0} \frac{2 \cos x}{\cos x (1 + 2 \sin x)} \\ & = \lim_{x \rightarrow 0} \frac{2}{1 + 2 \sin x} \\ & = \frac{2}{1 + 2 \cdot 0} = 2 \end{aligned}$$

References